

Project Overview

- This project aims to design a vinyl chloride monomer (VCM) recovery system for a PVC process.
- The primary goal is to optimize the yield of VCM while minimizing VCM loss to the incinerator.

Process Safety

Vinyl Chloride Monomer (VCM) Hazards

Toxicity: Known human carcinogen (TLV-TWA = 1 ppm)

Health Risk: Severe effects at higher concentrations (per ERPG guidelines)

Flammability:

- Flash point: -108°F
- Flammable range: 3.6–33 vol%
- Easily ignited under process conditions

Ignition Risk: Very low minimum ignition energy

Reactivity:

- Potential for runaway polymerization
- Explosive reactions with oxygen/oxidizers

Safety Controls Required:

- Stabilization
- Continuous monitoring
- Proper process control systems

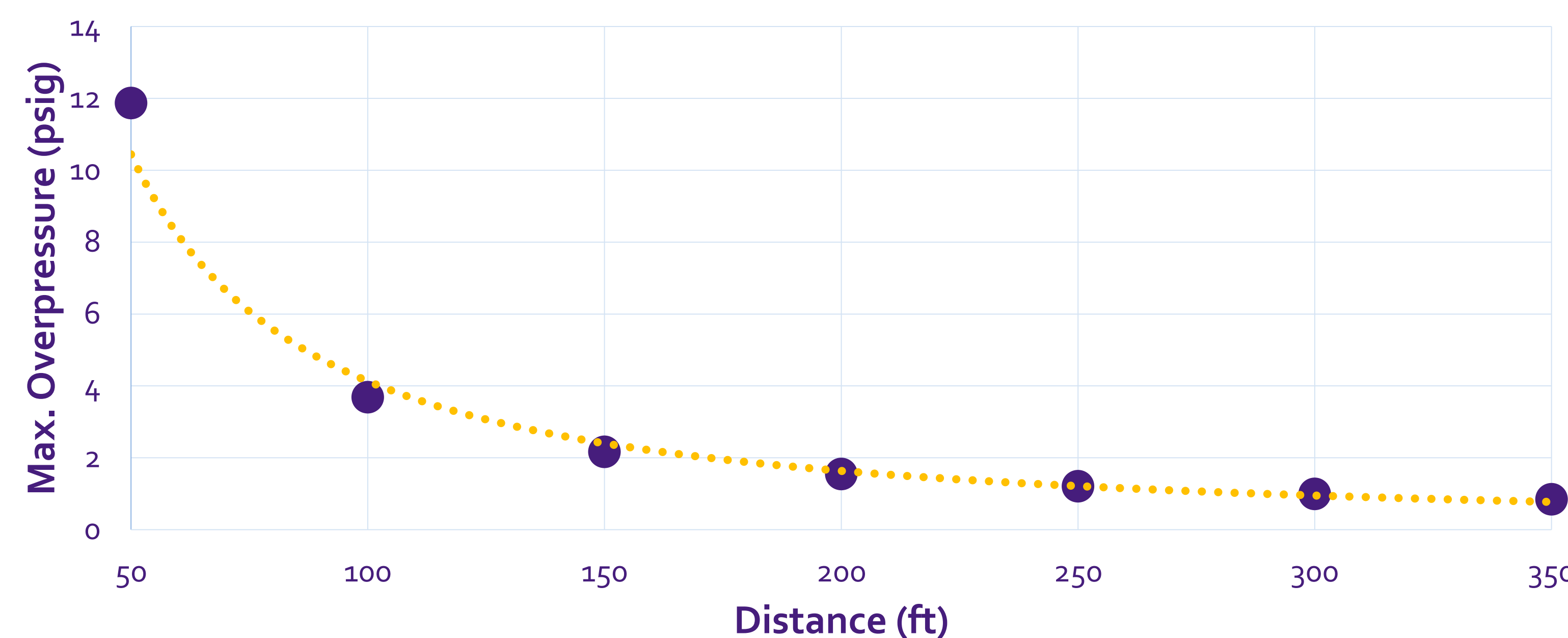
Consequence Assessment

Qualitative & Semi-Quantitative Risk Matrix

Adapted from Layers of Protection Analysis, CCPS, 2001, page 135 (semi quant) and CCPS PS for Engineers figure 14.5 page 321.

Initiating Event Frequency		Consequence Level				
Qualitative	Semi-Quantitative (yr ⁻¹)	1	2	3	4	5
Frequent	1.0E+00					
Likely	1.0E-01					A
Occasional	1.0E-02				B	
Unlikely	1.0E-03			C		
Rare	1.0E-04		D			
	1.0E-05					
	1.0E-06					

Maximum Overpressure vs. Distance



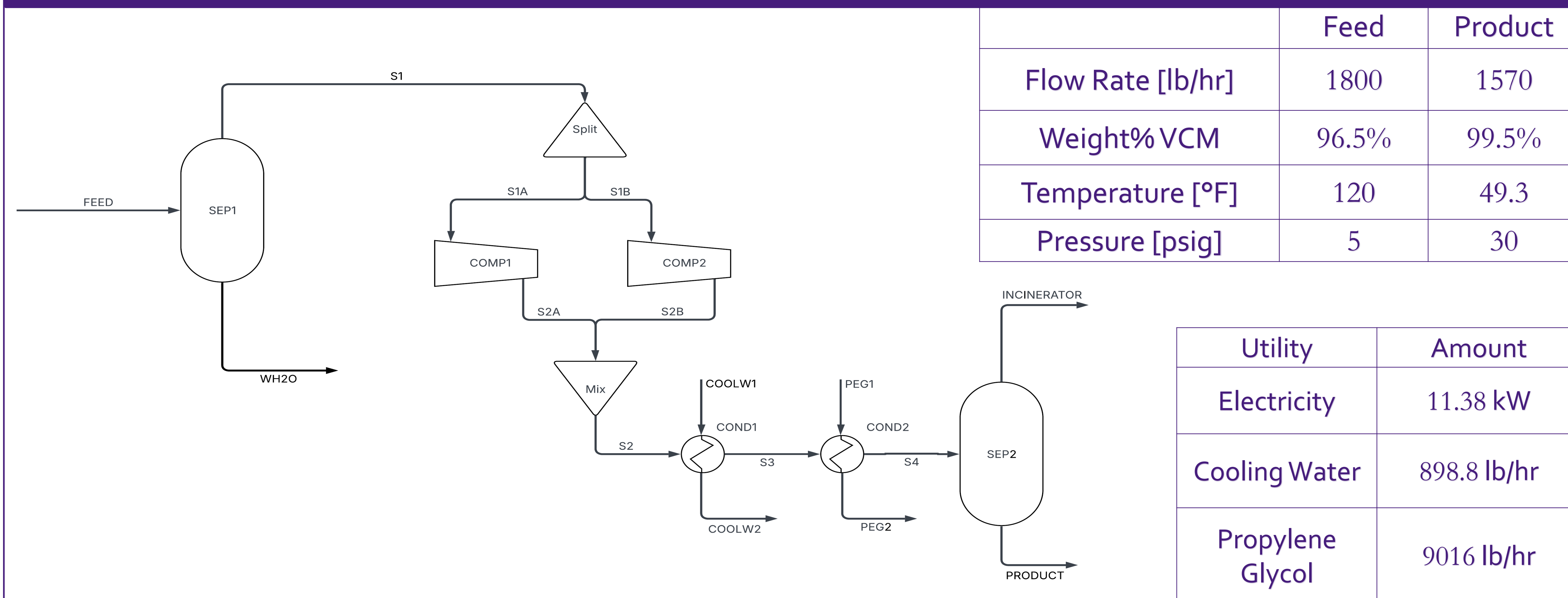
Economic Analysis

Total Capital Cost [USD/Year]	\$	3,713,450.00
Total Installed Cost [USD]	\$	1,967,500.00
Total Operating Cost [USD/Year]	\$	1,540,720.27
Total Utilities Cost [USD/Year]	\$	41,529.28
Total Product Sales [USD/Year]	\$	3,108,600.00
Return on Investment (ROI)		36.56%
Payback Period (PBP)		1.96 years

Equipment Cost Breakdown

Name	Category	Equipment Cost [USD]	Installed Cost [USD]	Utility Cost [USD/HR]
COND2	Condenser	13400	85600	0.459
COMP2	Compressor	687900	841800	0.324
COND1	Condenser	10500	59200	0.024
SEP1	Separator	22100	139100	0
COMP1	Compressor	687900	841800	0.324

Process Flow Diagram



Final Recommendations

Exceptional Economic and Operational Performance:

The proposed VCM recovery system is highly recommended for implementation, as it successfully captures 90% of unreacted vapor (yielding 1570 lb/hr at 99.5% purity) while delivering a highly profitable 36.56% ROI and a 1.96-year payback period on a \$2.3MM capital investment.

Safe Operability with Cost-Reduction Potential:

The design maintains an acceptable "C" risk ranking through the required use of robust safeguards (IPLs) and strict maintenance protocols, while offering a clear pathway for future optimization—such as transitioning to a single multi-stage compressor—to potentially decrease total installed costs by 43%.